Microsoft

Q A Maker

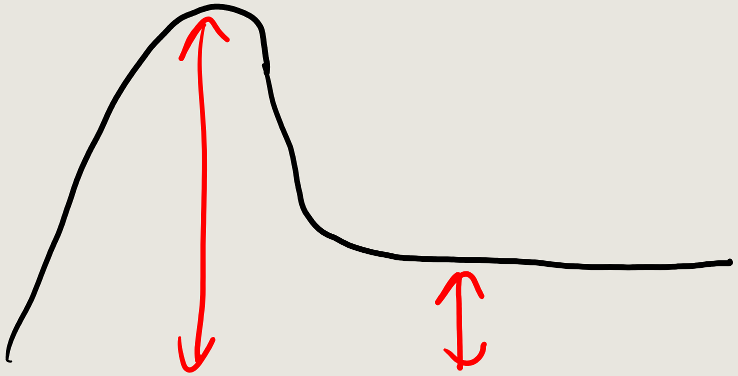
Bot

AI-103

Developing Azure AI Apps and Agents

https://aka.ms/mslearn-develop-ai-agents

Agenda



GenAI
LLMs

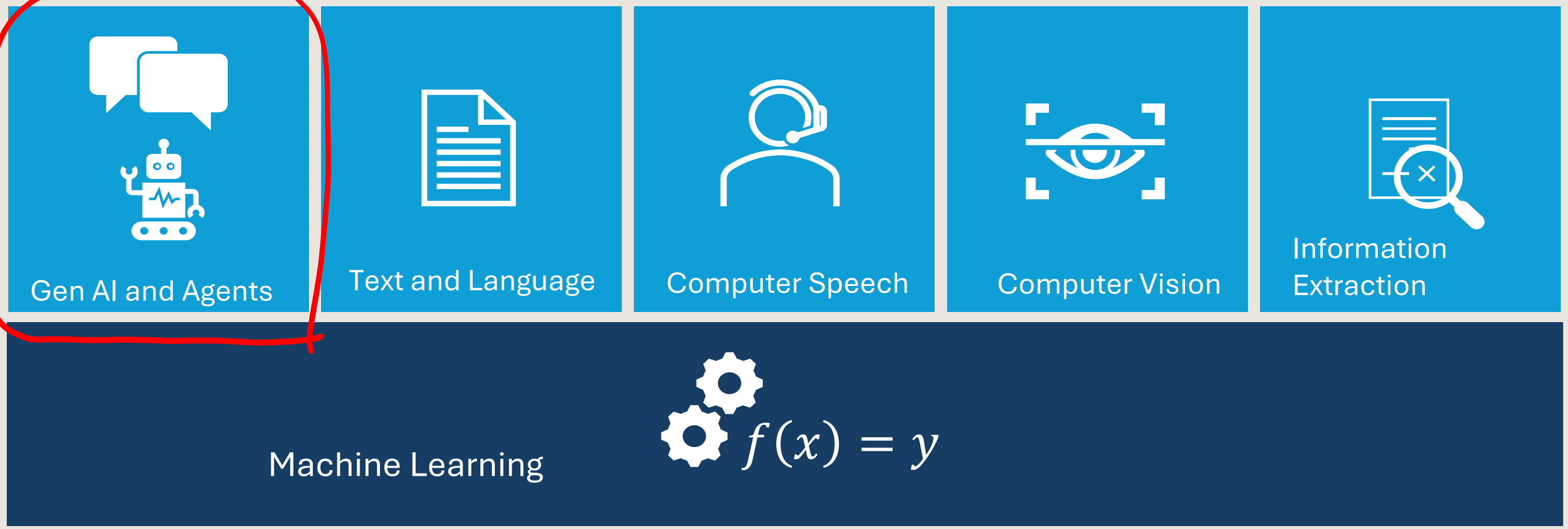
1. Develop generative AI apps in Azure
2. Develop AI agents in Azure

Classic

3. Develop natural language solutions in Azure

NLP

4. Extract insights from visual data on Azure



Our focus today

Generative AI and Agents

Text and Language

Computer Speech

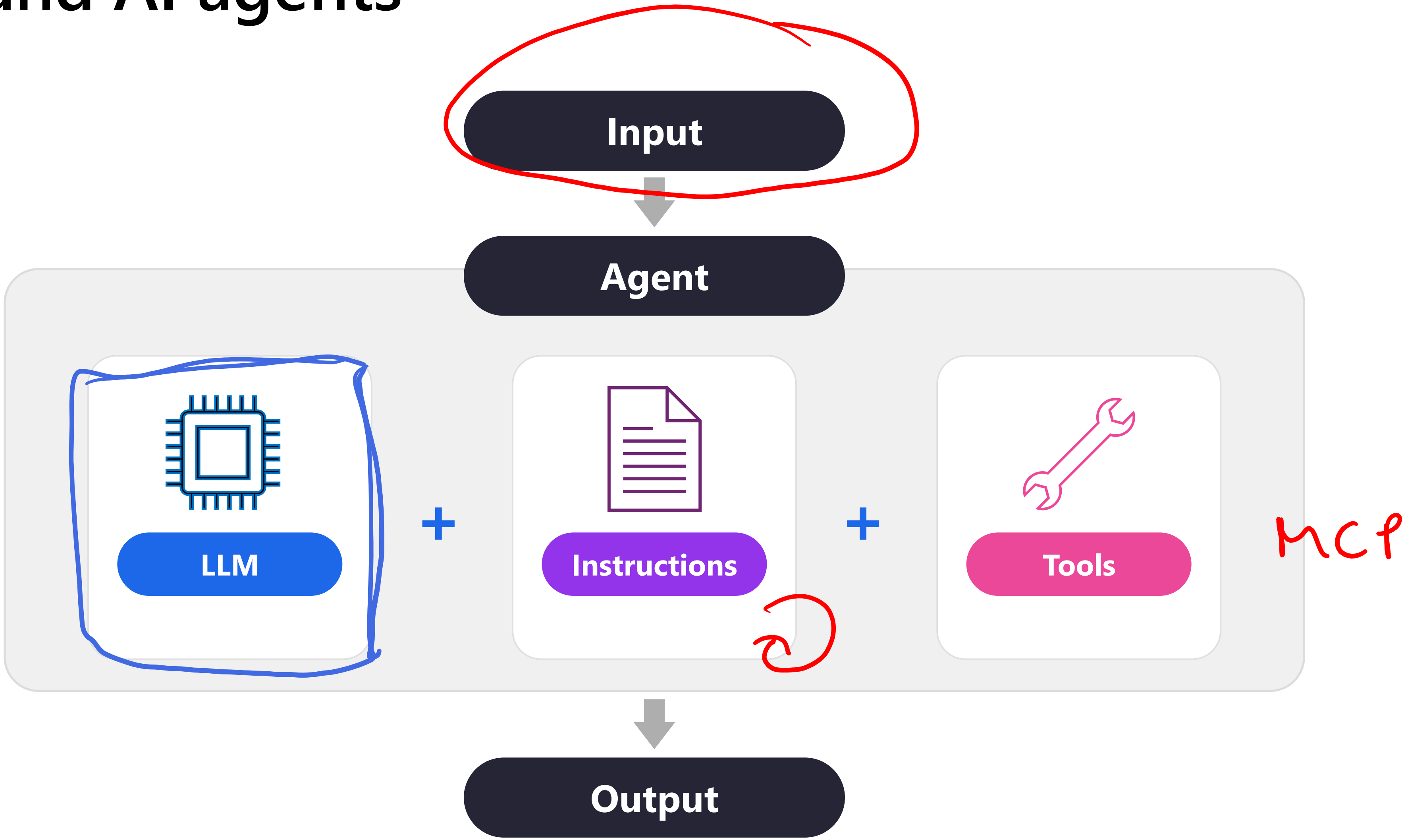
Computer Vision

Information Extraction

$f(x) = y$

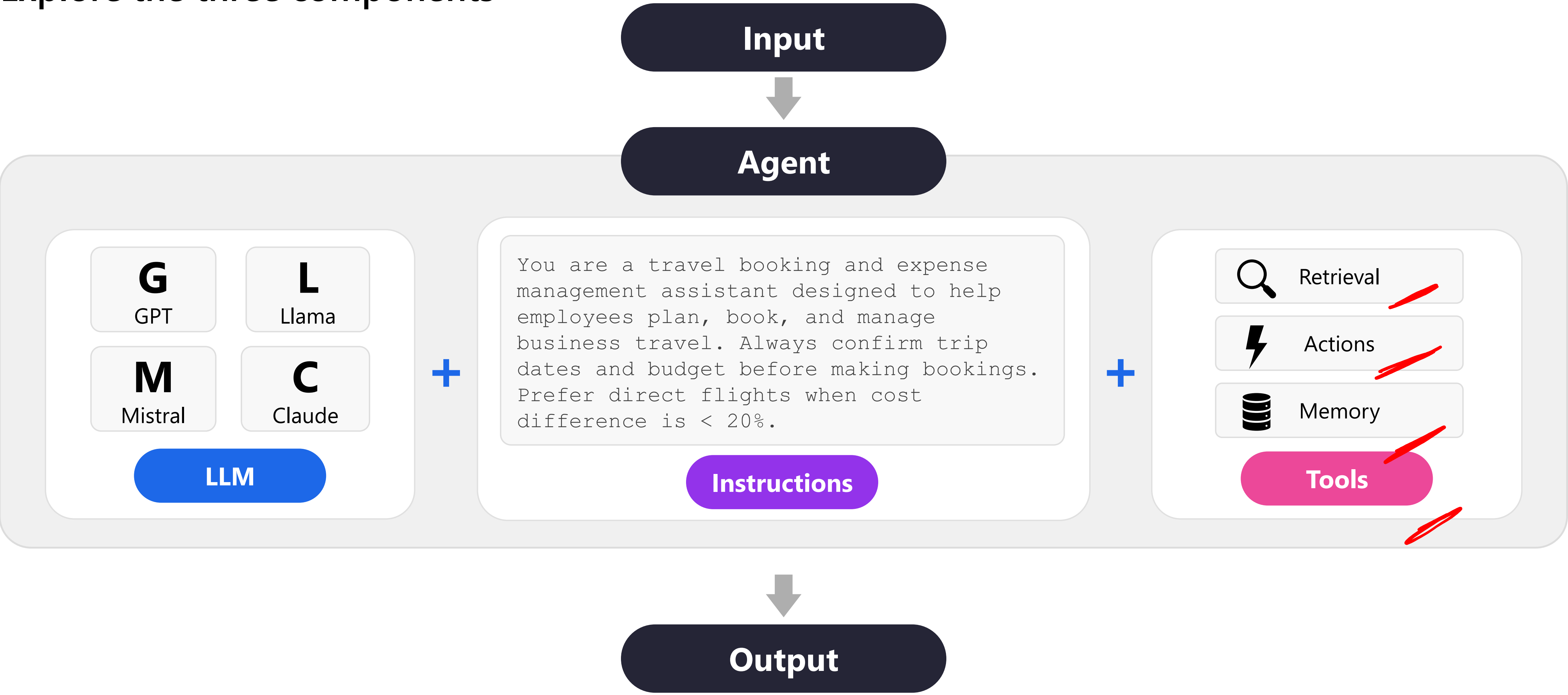
Machine Learning

Understand AI agents



Understand AI agents

Explore the three components



Why agents are useful

Automation of routine tasks
Handles repetitive activities, freeing humans to focus on strategic and creative work.

Enhanced decision-making
Analyzes data to surface insights, predict outcomes, and recommend actions in real time.

Agent

Scalability
Grows capabilities without proportional increases in headcount or operational costs.

24/7 availability
Operates continuously without breaks, keeping services available around the clock.

AI agent use cases

Personal productivity

Schedule our team standup for next week.

Done! Mon–Fri at 9 AM — invites sent to all 6 team members.

Also draft a recap email from yesterday's notes.

Draft ready in your Drafts folder. Want me to send it?

Research

Any major shifts in the AI market this week?

3 key updates: AI adoption up 40%, EU AI Act enforcement started, healthcare AI funding at record high.

Generate a summary report for the team.

Report generated — 2-page summary with source links attached.

Sales

Find leads in the Pacific Northwest retail sector.

Found 14 leads. Drafted personalized emails and flagged 3 as high-priority for calls.

Schedule the priority calls for this week.

Calls booked Tuesday and Thursday — calendar invites sent.

Customer service

I need a refund for order #4821.

Refund approved! $47.99 will return to your card in 2–3 business days.

Can I get a confirmation email?

Sent to your registered email. Is there anything else I can help with?

Typical development workflow

1. Connect to your Foundry project
2. Create an AI agent
3. Configure agent instructions
4. Add tools
5. Test the agent in playground
6. Iterate on design
7. Deploy the agent to production
8. Integrate into applications

Required resources

- **Microsoft Foundry project**
  Organizes agents, models, and assets
- **Model deployment**
  For example, GPT-4.1 or Claude Sonnet 4.6

Optional resources

- **Azure AI Search** — Knowledge retrieval for Foundry IQ or File Search
- **Azure Storage** — Files agents can read and write
- **Azure Key Vault** — Securely store secrets and credentials
- **Azure Functions** — Custom tool implementations and business logic

Erich Gamma

Microsoft

Choose your development approach

Foundry portal

Visual, web-based, no local setup required

- ✓ **Quick prototyping**
  Test agent concepts without setting up a local development environment
- ✓ **Visual configuration**
  Configure agents through intuitive forms and dropdowns rather than code
- ✓ **Centralized management**
  View and manage all agents across projects in one place with dashboards
- ✓ **Team collaboration**
  Share configurations with stakeholders who prefer visual interfaces

OR

Visual Studio Code

Developer-centric, code-integrated, Git-friendly

- ✓ **Developer-centric workflows**
  Build agents alongside application code in a single familiar environment
- ✓ **Version control integration**
  Track agent configurations in Git alongside your codebase
- ✓ **Code-first development**
  Edit YAML configurations directly and integrate agents into application logic
- ✓ **Local development**
  Design and test agents offline before deploying to Azure

Many teams use both: the Foundry portal for initial exploration and reviews, and VS Code for detailed development and production deployments.

Git   Open Source   Linux

GitHub   Microsoft

Demo

Build AI agents with portal and VS Code

In this demo, you'll explore how to:

- Create an AI agent in Microsoft Foundry portal
- Interact with your agent using VS Code

Start the exercise at:

5. Build AI agents with portal and VS Code

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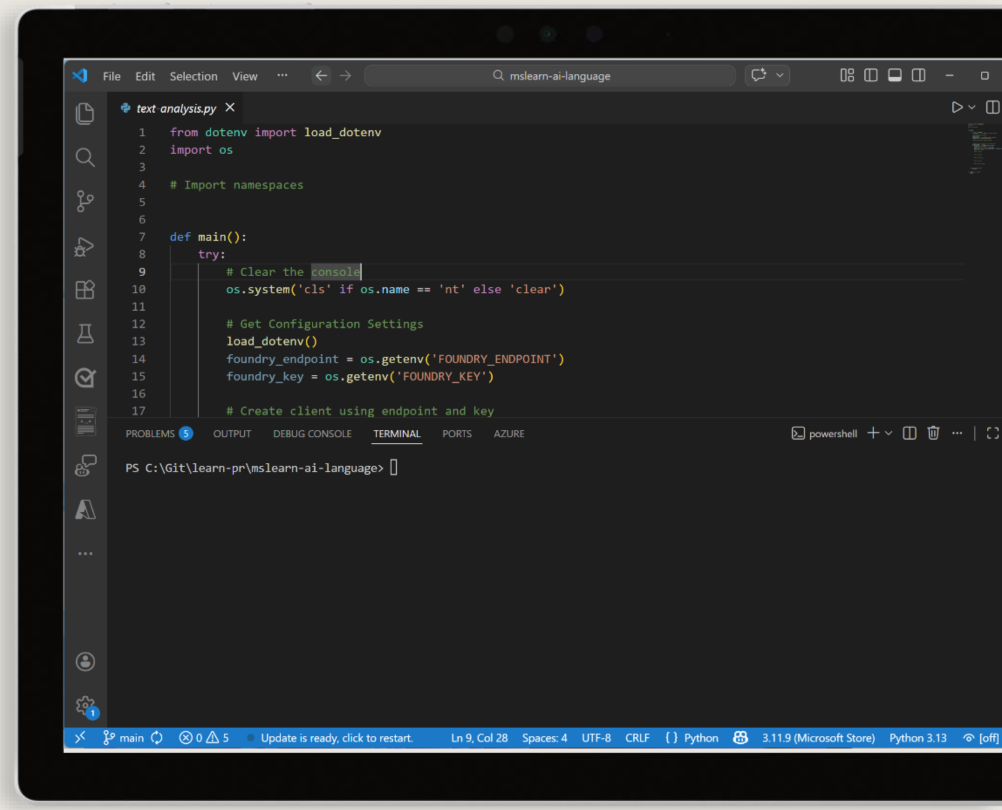


Knowledge check

1 **What is the primary benefit of using Microsoft Foundry Agent Service compared to building agents with standard APIs?**

☐ It provides access to more powerful AI models

☐ It requires no Azure subscription

☑ It handles tool calling, state management, and infrastructure automatically

☐ It only works with the Azure portal

2 **How does Microsoft Foundry Agent Service handle conversation state?**

☐ By requiring developers to manually manage conversation history

☐ Through external database connections

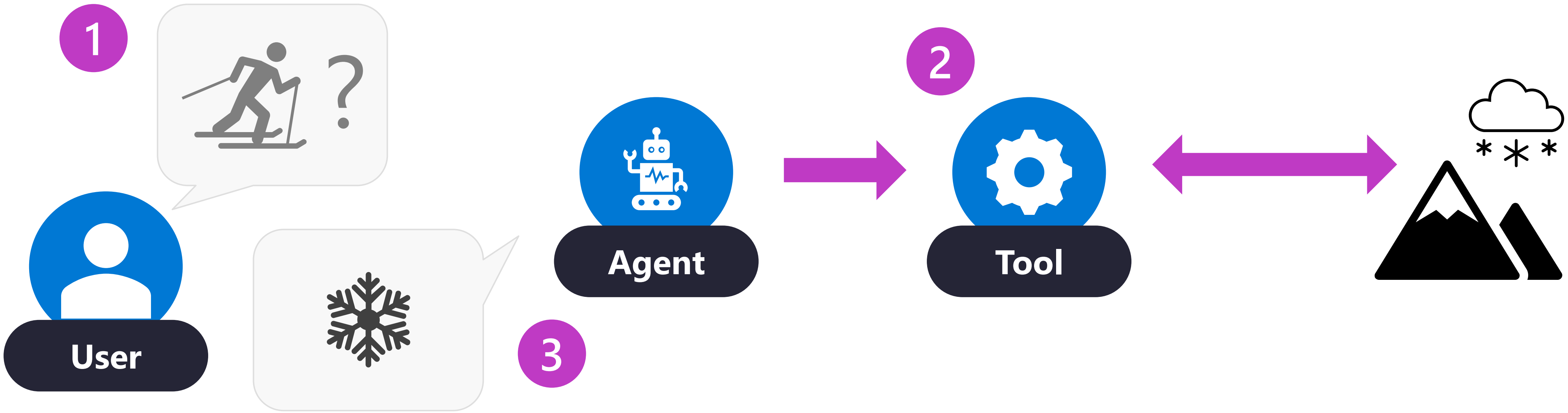
☑ Through the Responses API which automatically manages conversation context

☐ Using local file storage on the client device

Integrate custom tools into your agent

https://aka.ms/mslearn-agent-custom-tools

Why use custom tools?



1. User asks an agent about the weather conditions in a ski resort.
2. The agent determines it has access to a tool that can use an API to get meteorological information and calls it.
3. The tool returns the weather report, and the agent informs the user.

Options for implementing custom tools

N8N

No Code

Custom function

Describe custom functions to an agent and let it dynamically identify and call the right function with the correct arguments.

Supports multiple programming languages for integrating custom logic and workflows.

Function calling

Azure Functions

Build intelligent, event-driven applications with minimal overhead.

Triggers determine when a function executes; bindings streamline connections to input and output data sources.

Event-driven

OpenAPI specification

Build intelligent, event-driven applications with minimal overhead.

Triggers determine when a function executes; bindings streamline connections to input and output data sources.

OpenAPI 3.0

Azure Logic Apps

Build intelligent, event-driven applications with minimal overhead.

Workflow

Triggers determine when a function executes; bindings streamline connections to input and output data sources.

Low-code

Exercise

Build an agent with custom tools

In this exercise, you'll:

- Create a function for the agent to use
- Define the function tools
- Create the agent that uses the function tools
- Send a message to the agent and process the response
- Process function calls and display the agent's response

Start the exercise at:

6. Use a custom function in an AI agent

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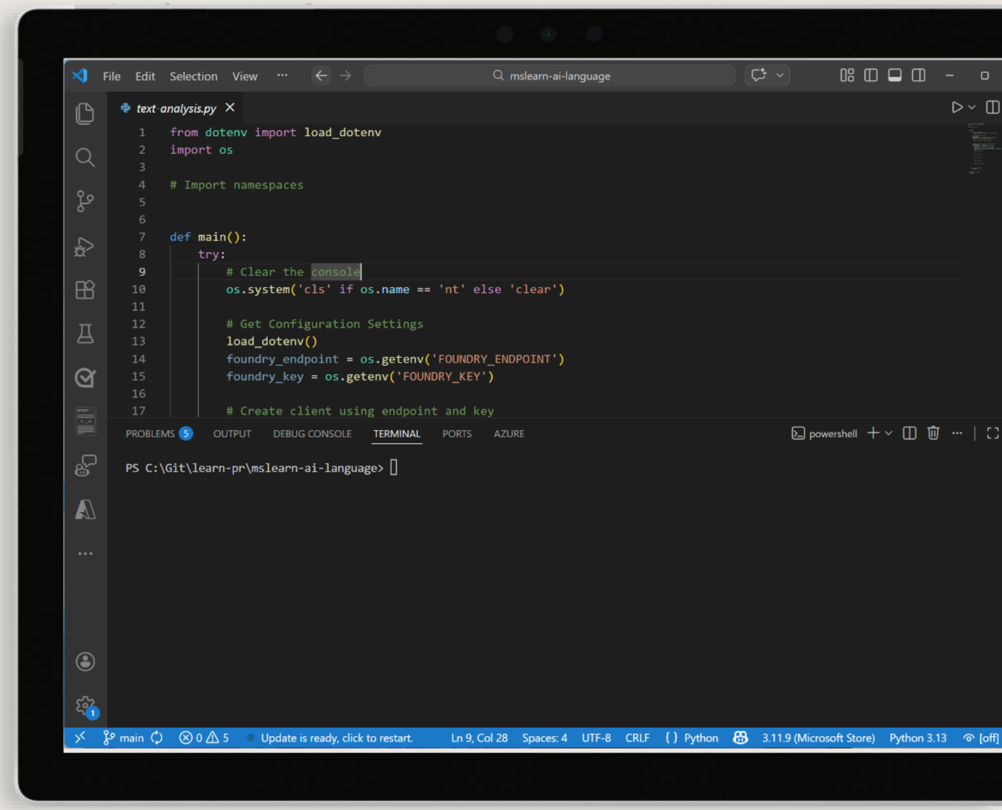


Knowledge check

1 **What are custom tools, and how can they help you develop effective agents?**

☑ Callable functions that an agent can use to extend its capabilities

☐ Extensions for Visual Studio Code to create and deploy agents

☐ Fine-tuned models that the agent uses to generate custom output

2 **You need to integrate functionality from an OpenAPI 3.0-based web service. What should you do?**

☐ Add the JSON schema to the agent's instructions

☐ Rewrite the web service as a Python function and hard-code it

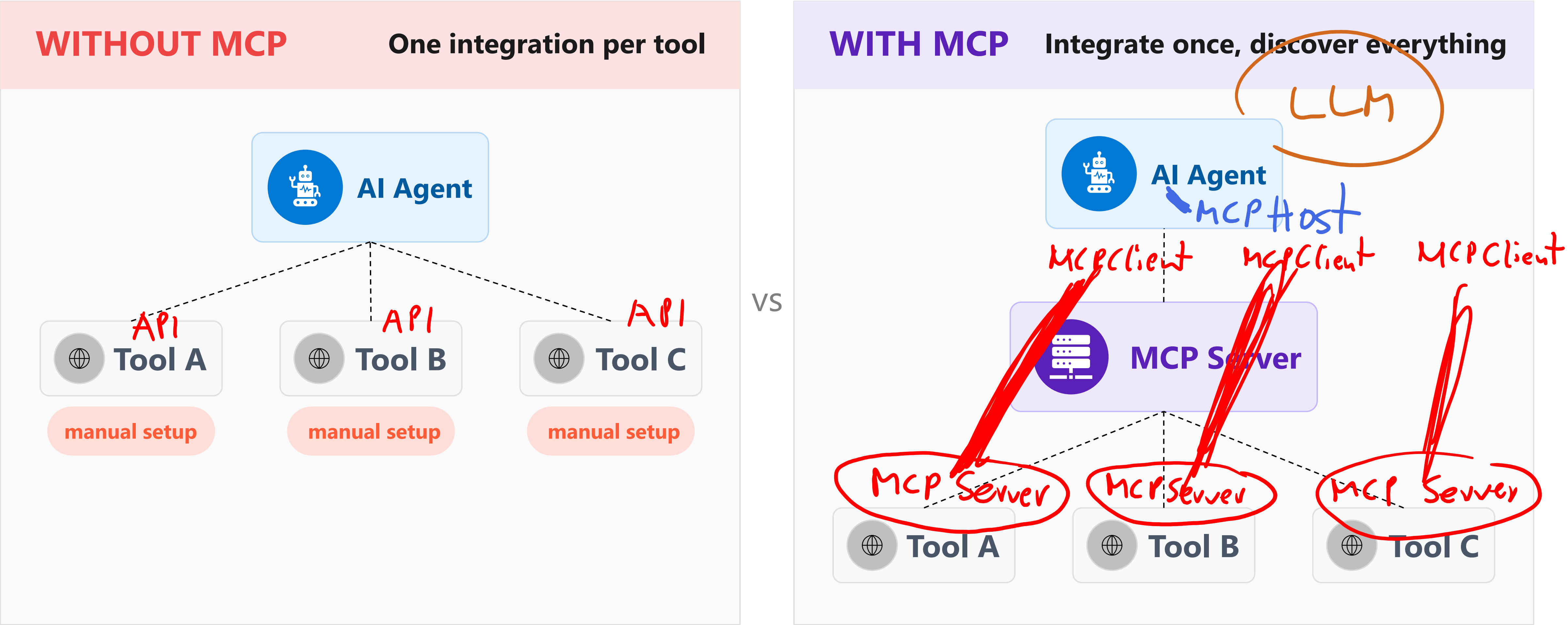
☑ Add the web service as an OpenAPI specification tool to the agent definition

Anthropic

Integrate MCP tools with Azure AI agents

https://aka.ms/mslearn-agent-mcp

Understand Model Context Protocol for AI agents



Integrate agent tools using an MCP server and client

MCP Server

Host tool definitions

`@mcp.tool`

MCP Client

Initialize session

`ClientSession(...)`

Fetch available tools

`session.list_tools()`

Wrap each tool

`session.call_tool`

Agent

Register to agent's toolset

`FunctionTool`

Exercise

Connect MCP tools to Foundry Agents

In this exercise, you'll:
- Connect a Foundry Agent to a remote MCP server
- Connect a Foundry Agent to custom MCP server tools

Start the exercise at:

7. Develop an AI agent with Model Context Protocol (MCP) tools

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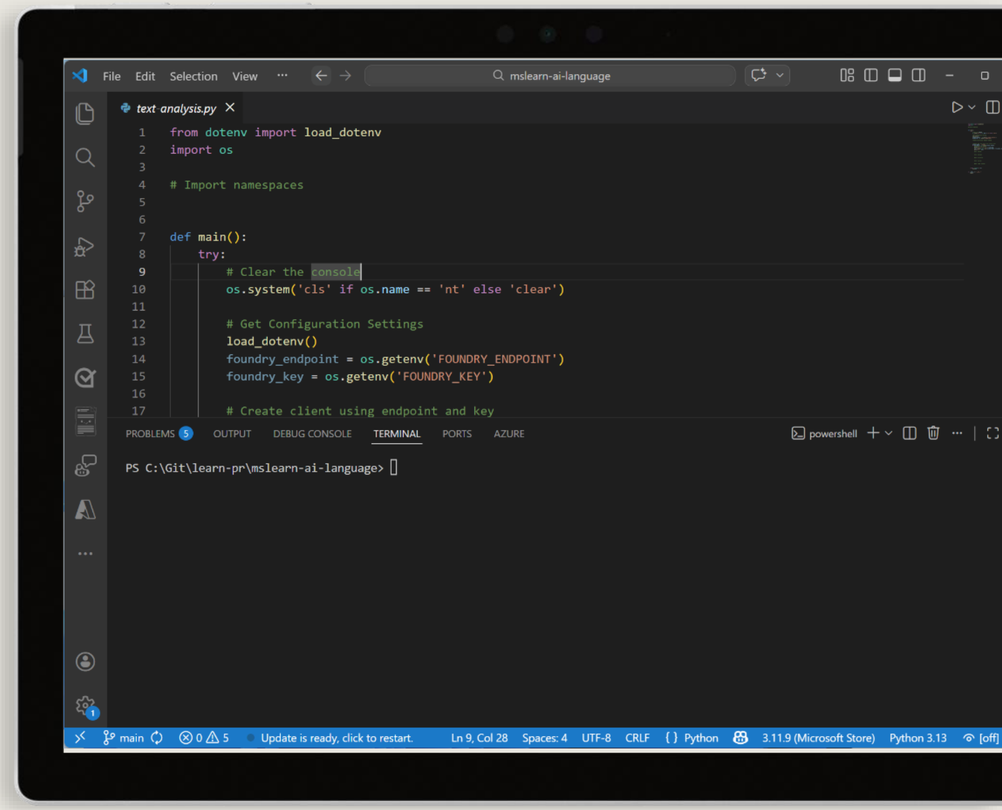


Knowledge check

1 What role does the MCP server play in the MCP agent tool integration?

- ☐ Runs the AI agent and processes user prompts directly
- ☐ Manages network connections between multiple agents
- ☑ Hosts tool definitions and makes them available for discovery by the client

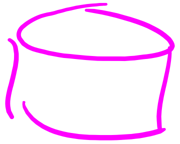
2 How does an MCP client retrieve available tools from the MCP server?

- ☑ By calling session.list_tools() to get the current tool catalog
- ☐ By reading a static JSON file from the server directory
- ☐ By subscribing to server events via a WebSocket connection

Build knowledge-enhanced AI agents with Foundry IQ

https://aka.ms/mslearn-agent-foundry-iq

Retrieval Augmented Generation (RAG) for agents

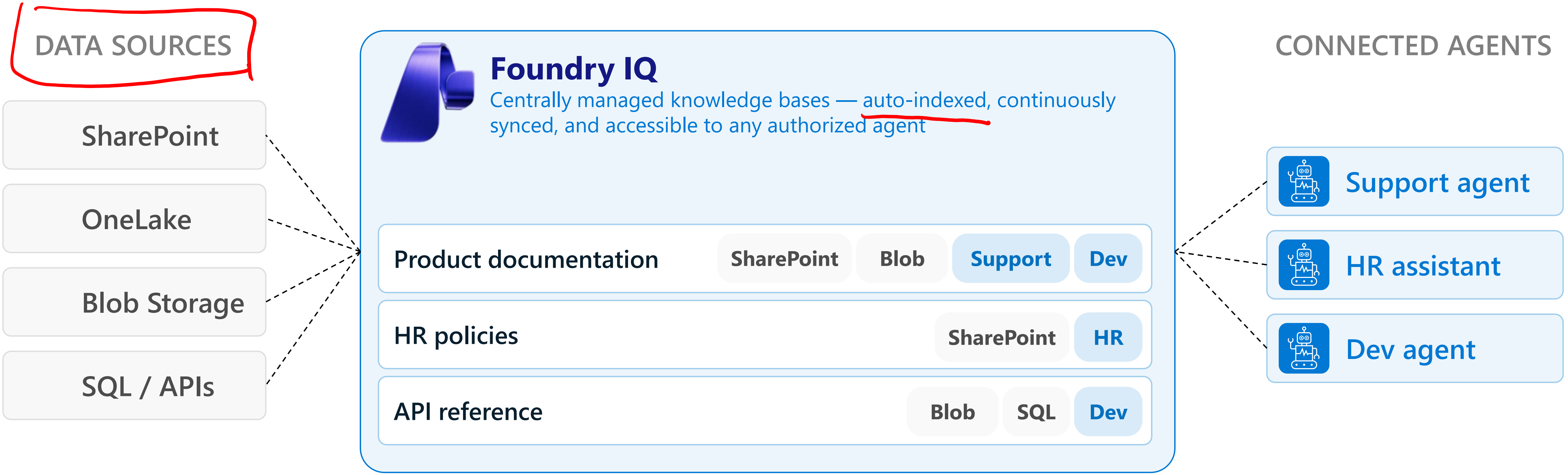


Vector DB

LIMITATION	Simple agent	RAG-powered agent
Knowledge cutoff	**Outdated answers** Unaware of anything past its training cutoff date	**Always current** Retrieves live documents from connected knowledge bases
No private data	**Generic only** No access to internal procedures or product specs	**Org-specific** Searches internal knowledge bases for company content
Lack of context	**Irrelevant advice** Ignores org workflows, approval chains, and security policies	**Contextually aware** Grounds advice in retrieved org policies and workflows
Fabricated responses	**Compliance risk** Confident-sounding answers with no source to verify	**Verifiable sources** Cites the exact document behind every response
Scalability	**Duplicated effort** Every team rebuilds the same RAG pipeline independently	**Shared platform** One knowledge platform (e.g., Foundry IQ) serves all teams

Explore Foundry IQ

A managed knowledge platform built into Microsoft Foundry
Define your knowledge bases once, connect any number of agents to them



IQ Foundry

Configure data sources

Indexed

Pre-processed | High relevance | Custom

Azure AI Search Index
BEST FOR
Enterprise search with custom ingestion pipelines and AI enrichment

SharePoint (Indexed)
BEST FOR
Advanced search on SharePoint content with custom query pipelines

Direct

No index needed | Raw file access

Azure Blob Storage
BEST FOR
Document files in Azure Storage — PDFs, DOCX, text, and more

OneLake
BEST FOR
Unstructured data and documents in Microsoft Fabric's unified data lake

Real-time

Always current | Live retrieval | No data copy

Web
BEST FOR
Current public information retrieved via Bing at query time

SharePoint (Remote)
BEST FOR
Live SharePoint content with Microsoft 365 governance and permissions

Configure retrieval with Foundry IQ

Effective instructions specify three critical behaviors

When to retrieve

Tell the agent to always use the knowledge base, never rely on training data

How to cite

Specify the exact format for source attribution

What to do when unsure

Define fallback behavior when information isn't found

```
retrieval_instructions = """You are a helpful HR assistant.

CRITICAL RULES:
-  You must ALWAYS search the knowledge base before answering
   any question
-  You must NEVER answer from your own knowledge or training
   data
-  Every answer MUST include citations
-  If the knowledge base doesn't contain the answer, respond
   with "I don't have that information in our current
   documentation. Please contact HR directly at
   hr@company.com"

Your role is to provide accurate, verifiable information from
company documentation."""
```

Exercise

Integrate an AI agent with Foundry IQ

In this exercise, you'll:

- Create an agent
- Configure your data and Foundry IQ
- Test the agent in the playground
- Connect to your agent from a client application
- Test the integration

Start the exercise at:

8. Integrate an AI agent with Foundry IQ

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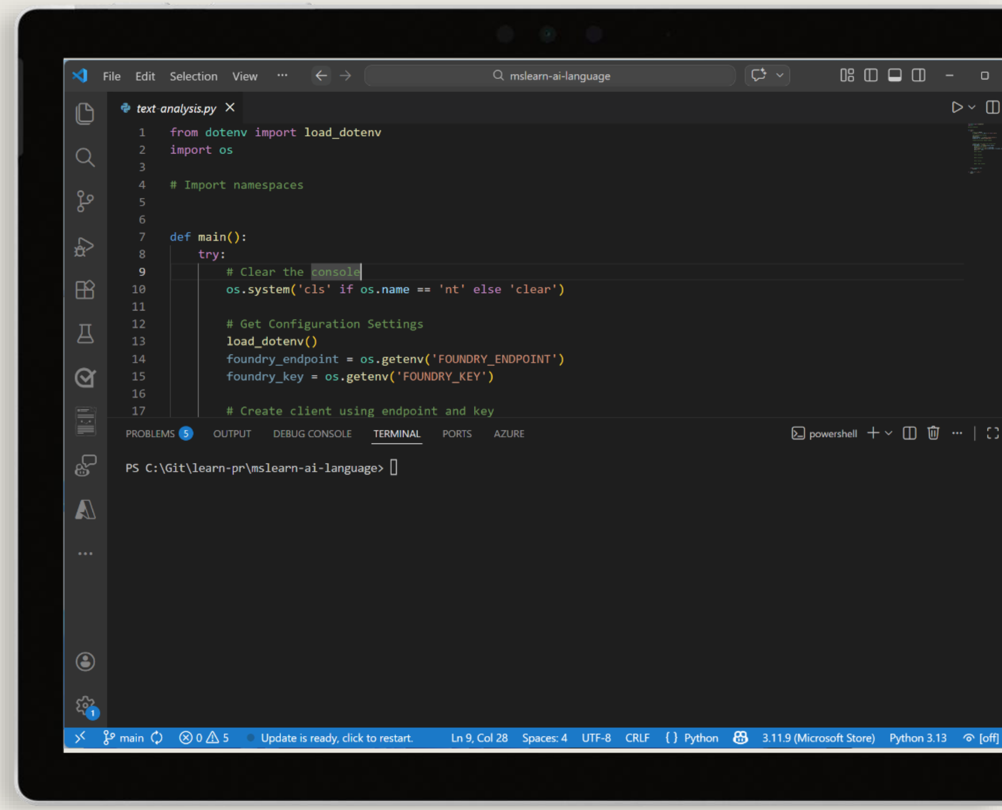


Knowledge check

1 **What is the primary advantage of Retrieval Augmented Generation (RAG) over simple AI agents?**

☐ RAG eliminates the need for large language models

☐ RAG enables agents to ground responses in current organizational information and provide source transparency

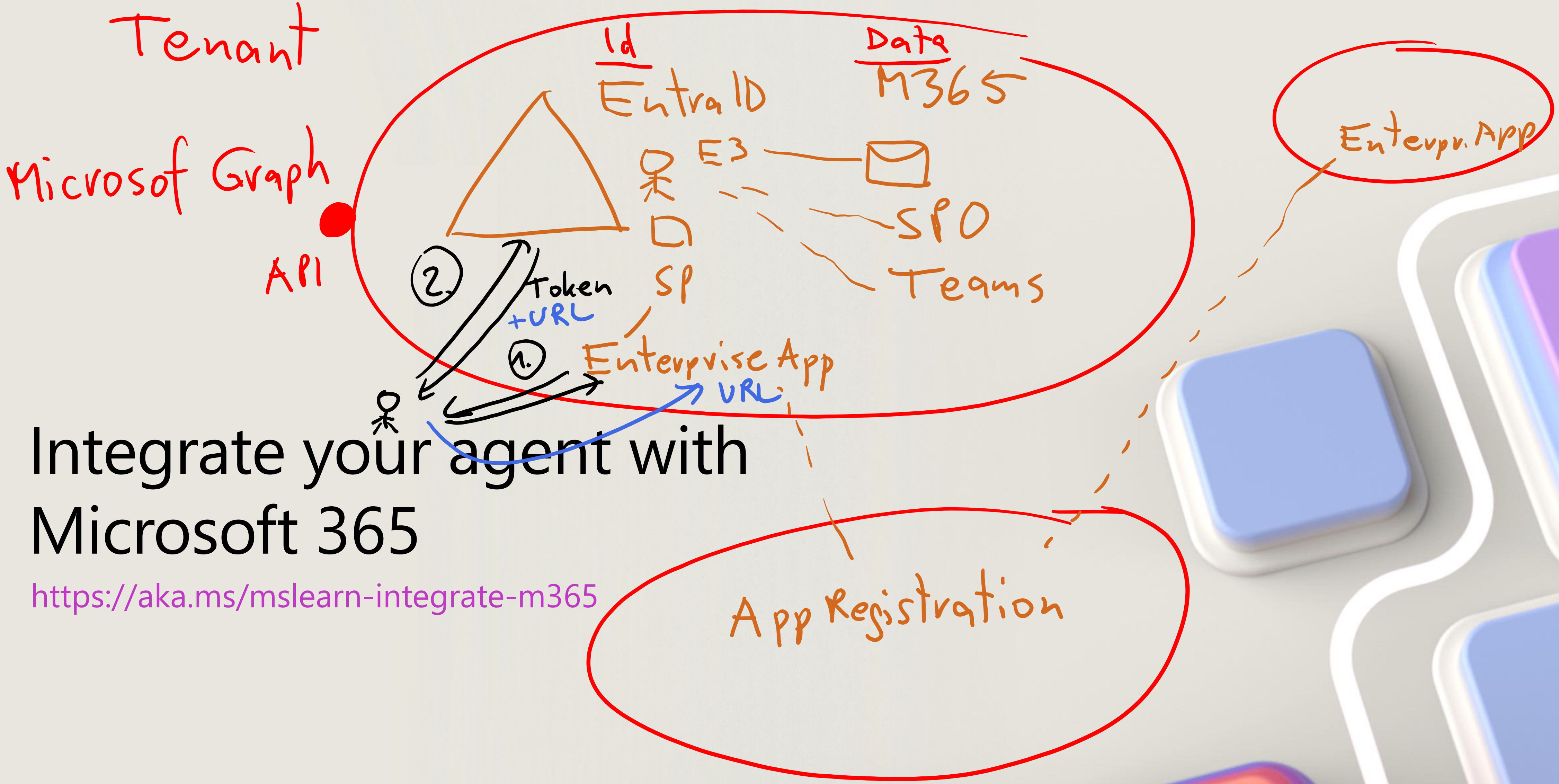
☐ RAG automatically retrains the language model when documents change

2 **Which data source option provides real-time access to SharePoint content with Microsoft 365 governance?**

☐ SharePoint Indexed, which pre-processes content into Azure AI Search

☐ SharePoint Remote, which queries SharePoint sites and libraries in real-time

☐ Azure Blob Storage, which connects to SharePoint files stored as blobs



Integrate your agent with Microsoft 365

https://aka.ms/mslearn-integrate-m365

Understanding agent applications

When you publish an agent, Microsoft Foundry creates an **Agent Application** resource with:

Dedicated invocation URL
A stable endpoint that remains consistent as you update agent versions.

Agent identity
A distinct Microsoft Entra identity separate from your development project

User data isolation
Inputs and interactions from one user aren't available to other users

Copilot Studio

Publish your agent

1 Select your agent version

2 Start the publishing process

3 Configure Azure Bot Service

4 Complete the **metadata**

metadata

Name	Display name for your agent in the Teams store
Description	Brief explanation of what your agent does
Icons	Upload small and large PNG icons
Publisher information	Your organization name and contact details
Privacy policy	URL to your organization's privacy policy
Terms of use	URL to your terms of service

5 Choose your publish **scope**

Shared scope
Agent appears under "Your agents" in the store.
Available immediately. Best for testing and small teams.

OR

Organization scope
Agent appears under "Built by your org" in the store.
Requires admin approval. Best for production deployments.

Tenant

6 Prepare and optionally download the package

Demo

Publish a Foundry agent to Teams

In this demo, you'll explore how to:

- Create an agent in the portal
- Add knowledge with File Search
- Publish to Microsoft Teams
- Publish to Microsoft 365 Copilot
- Update published agents

Start the exercise at:

9. Deploy agents to Microsoft Teams and Copilot

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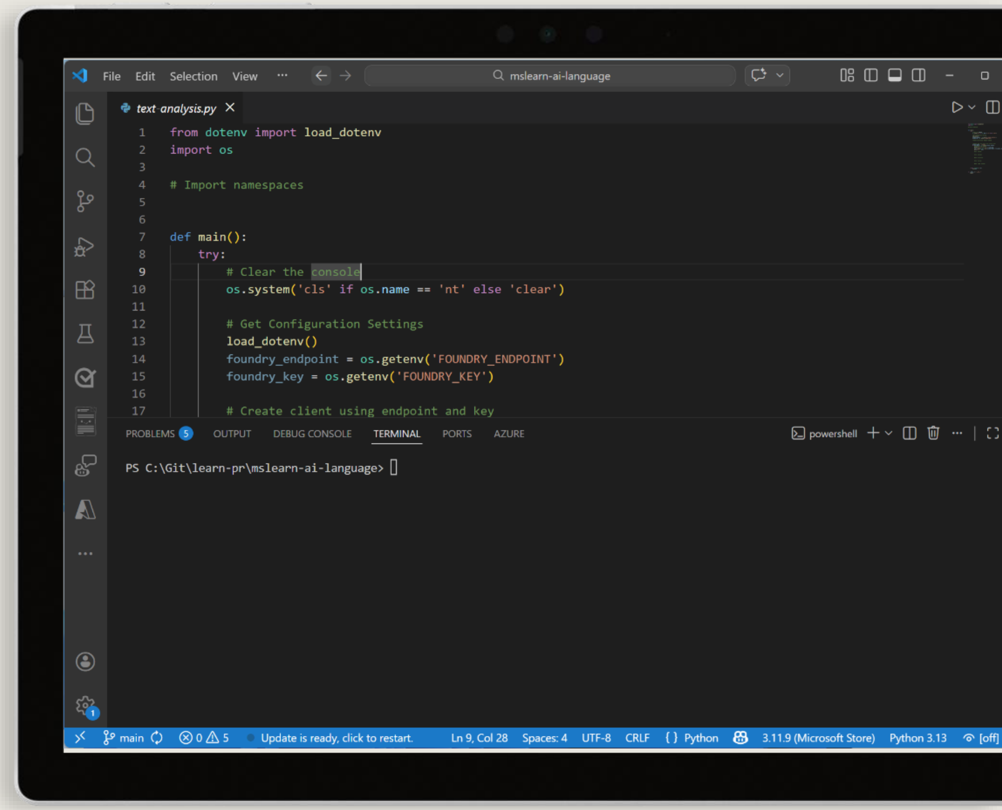


Knowledge check

1. **What Azure resource does the Foundry portal automatically create when you publish an agent to Microsoft Teams?**
   - ☐ Azure Functions
   - ☑ Azure Bot Service
   - ☐ Azure Cosmos DB
   - ☐ Azure Logic Apps

2. **What is Microsoft Work IQ?**
   - ☐ A machine learning model for workplace analytics
   - ☑ A CLI and MCP server that connects AI agents to Microsoft 365 data
   - ☐ A replacement for Microsoft Teams
   - ☐ A Visual Studio Code extension for building agents

Azure: Logic App → json

N8N: Workflows

Build agent-driven workflows using Microsoft Foundry

https://aka.ms/mslearn-agent-workflows

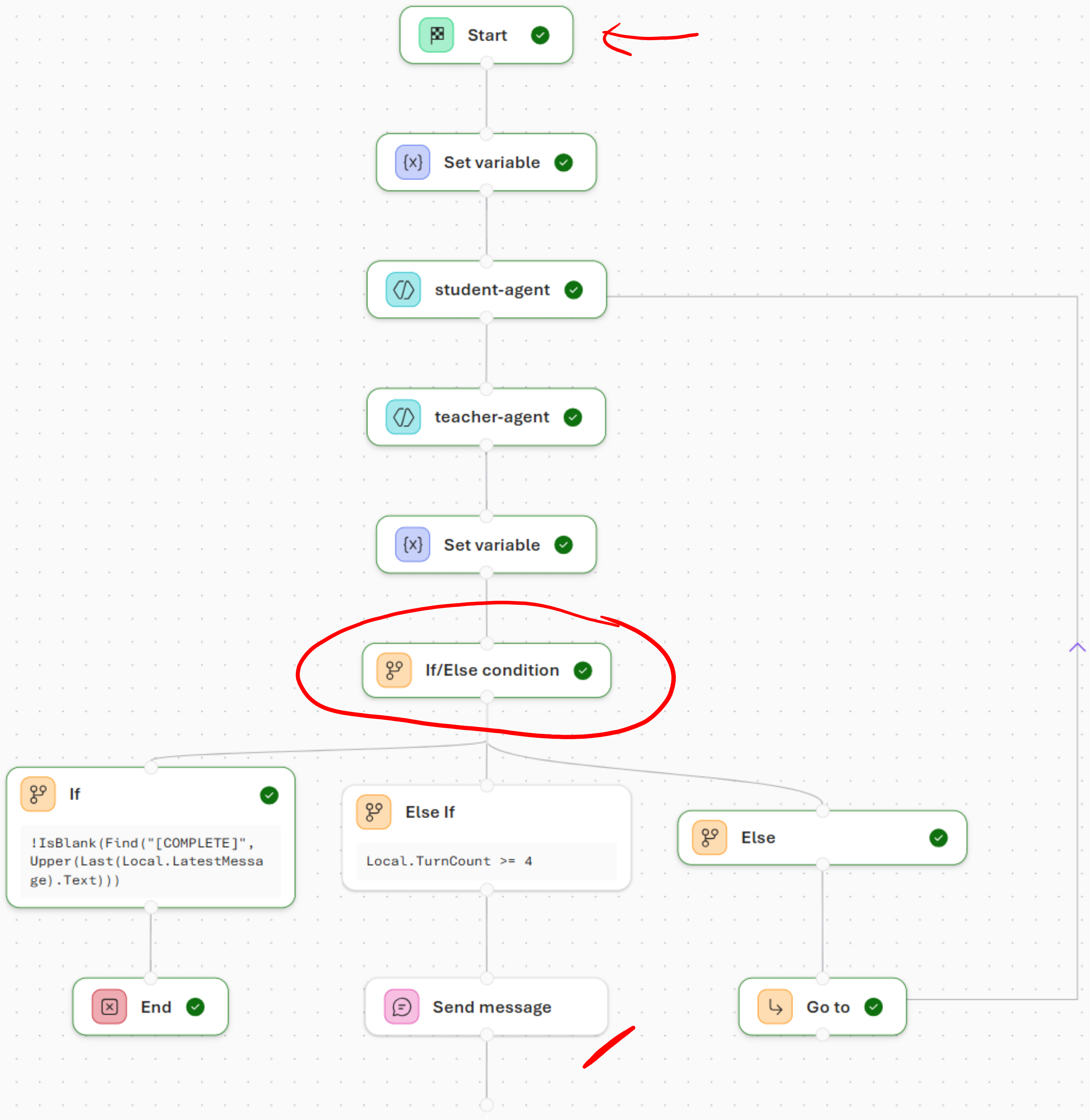
Understand workflows

Orchestrate AI-driven actions

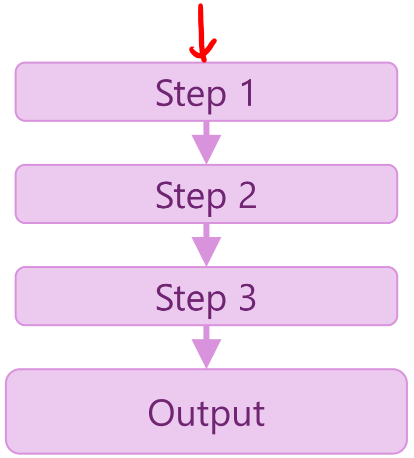
Rather than writing code, you define a sequence of steps using a visual, declarative approach that describe what should happen and when, allowing the platform to manage execution and state.

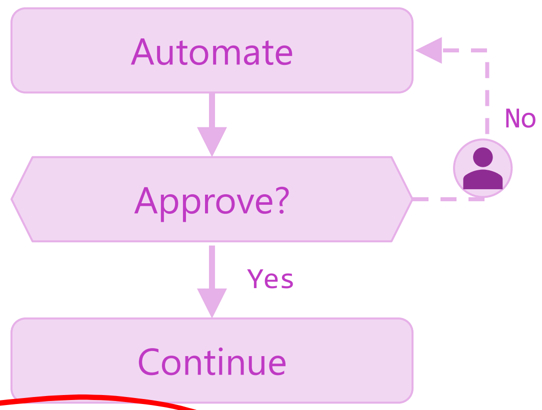
Connect the nodes

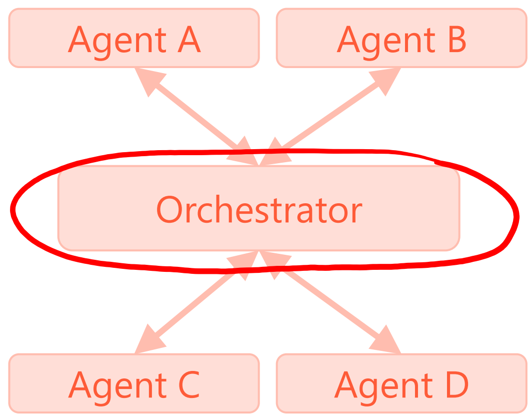
Each node performs a specific function. Some nodes invoke agents, while others evaluate conditions, manage data, or communicate with users. Together, these nodes form an execution path that determines how requests move through the system.



Identify workflow patterns







Sequential

Steps execute in a fixed order, each passing its output to the next. Best for predictable multi-stage pipelines.

Human-in-the-loop

Workflow pauses for human approval or input before continuing. Balances automation with oversight.

Group chat

Multiple specialized agents collaborate dynamically. Control shifts based on context, allowing agents to build on each other's outputs.

Explore core components of a workflow

Executors

The main workers in a workflow. Each executor receives input, performs actions, and passes outputs along — driving the workflow toward its goal.
Executors can represent AI agents or custom logic components.

Edges

Direct
One executor feeds directly into the next in sequence.

Conditional ?
Triggers only when a condition is met; otherwise skipped.

Switch-case
Routes to different executors based on predefined conditions

Fan-out
Sends a single message to multiple executors simultaneously.

Fan-in
Combines messages from multiple executors into one final step

Events

- `WorkflowStartedEvent`
  Triggered when workflow execution begins.
- `WorkflowOutputEvent`
  Emitted when the workflow produces an output.
- `WorkflowErrorEvent`
  Occurs when an error is encountered during execution.
- `ExecutorInvokeEvent`
  Fired when an executor starts processing a task.
- `ExecutorCompleteEvent`
  Fired when an executor finishes its work.
- `RequestInfoEvent`
  Logged when an external request is issued.

Demo

Build a workflow in Microsoft Foundry

In this demo, you'll explore how to:

- Create a customer support triage workflow
- Preview the workflow
- Use your workflow in code

Start the exercise at:

11

~~10~~. Build a workflow in Microsoft Foundry

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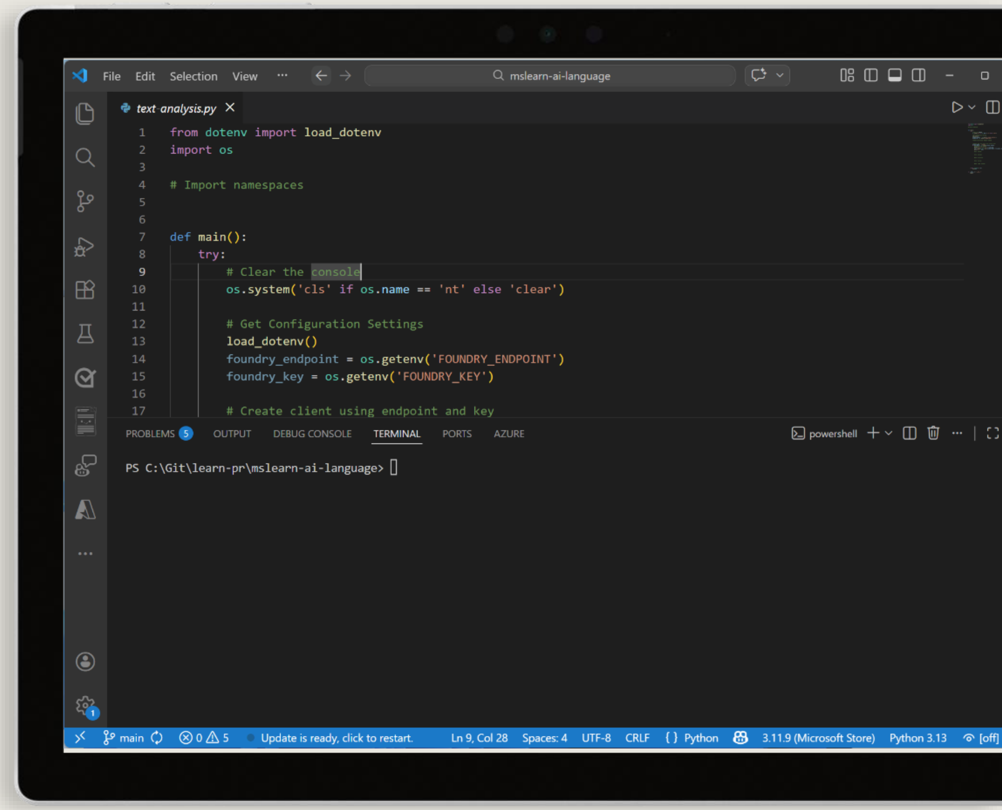

Knowledge check

1 Which type of node in a Foundry workflow is used to invoke an AI agent?

- [ ] Logic node
- [x] Agent node
- [ ] Data transformation node

2 Which node type would you use to handle multiple items in a workflow without duplicating nodes?

- [ ] If/Else node
- [x] For-Each node
- [ ] Send message node

Develop an AI agent with Microsoft Agent Framework

https://aka.ms/mslearn-agent-framework

Explore Microsoft Agent Framework core components

Agents
Consistent interface enabling multi-agent orchestration. Supports function calling, multi-turn conversations, structured outputs, and streaming responses out of the box.

Chat providers
Abstractions for connecting to AI services under a common interface. All providers are interchangeable through the BaseAgent abstraction.

Function tools
Containers for custom functions that extend agent capabilities. Agents automatically discover and invoke functions to integrate with external APIs and services.

Built-in tools
Prebuilt capabilities provided by the service, ready to use without additional configuration.

Conversation management
Structured message system with typed roles and AgentSession for persistent conversation context across multiple interactions.

Workflow orchestration
Patterns for complex multi-agent collaboration including human-in-the-loop support. Agents can hand off tasks or participate in group conversations.

Create an AI agent

1. Create a Microsoft Foundry project
2. Add the project connection string to your Microsoft Agent Framework application code
3. Set up authentication credentials with `AzureCliCredential`
4. Connect to your project client with the `AzureOpenAIResponsesClient` class
5. Create an `Agent` instance with the client, instructions, and tools you want to use

RBAC
EntraID
Global Admin.
Agent
SP
OAuth2.0
Access Token
RBAC

Terraform
Bicep — ARM
API

Azure
Foundry Resource
Project1
Project2

Foundry
Project1
2

Access Token

- models
- Guardrails
- Endpoints
- Tools
  code interpr.
- MCP-Server

Role
User
Access
Admin

RBAC
Role
Owner
Actions *
DataActions [ ]

Contributor
Action * aber nicht
Permission
DataActions [ ]

AI Owner
Cognitive Services *
Cognitive S *

Add tools to AI agents

Built-in tools

Code interpreter
Executes Python code for calculations, data analysis, and more

File search
Searches through and analyzes documents

Web search
Retrieves information from the internet

Custom function tools

Create a custom function tool by defining a Python function and decorating it with the `@tool` decorator

Create your tool by defining a regular Python function with proper type annotations

Pass your custom functions to the `ChatAgent` during creation using the `tools` parameter

Once tools are registered, the AI agent automatically selects and orchestrates the appropriate tools based on user requests, without requiring manual invocation.

Exercise

Develop an Azure AI agent

In this exercise, you'll:
- Install the Microsoft AI Toolkit VS Code extension
- Deploy a model
- Clone the starter code repository
- Write code for an agent app
- Run the app

Start the exercise at:

12

~~11~~. Develop an Azure AI chat agent with the Microsoft Agent Framework SDK

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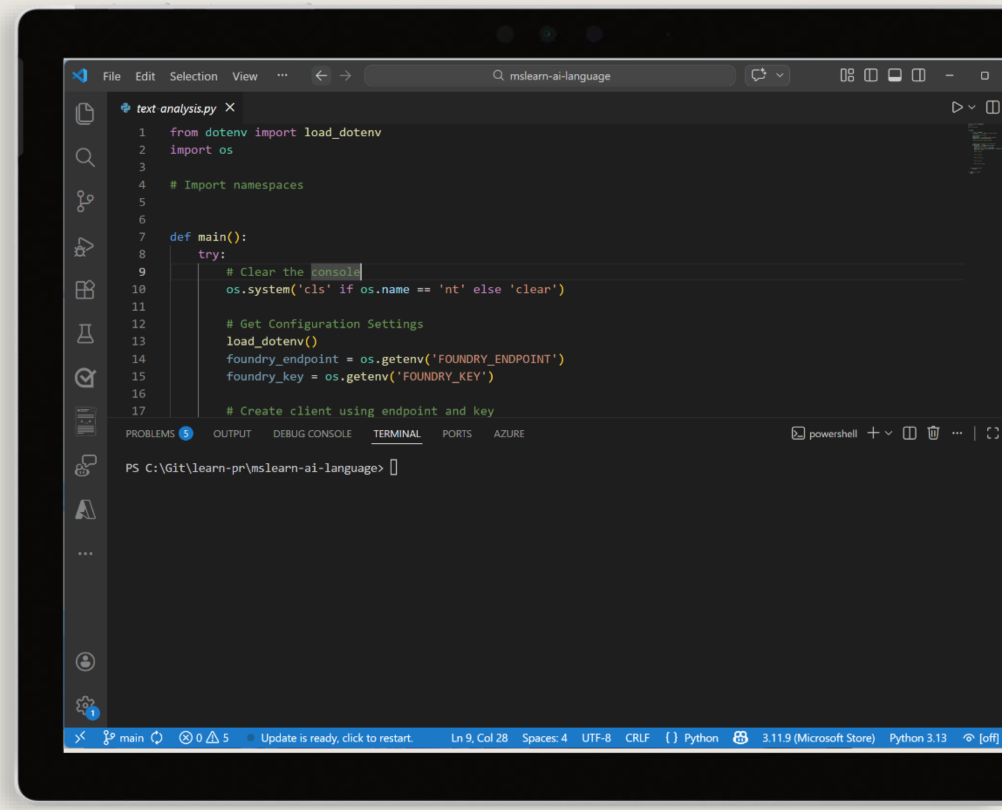


Knowledge check

1 **What are the key steps to create a Microsoft Foundry Agent using the Microsoft Agent Framework?**

☐ Deploy a custom AI model before creating an agent definition in the Azure portal

☐ Initialize the agent by defining a model in the `AgentThread` constructor

☑ Create an `AzureAIAgentClient`, define a `ChatAgent` with instructions and tools, and create an `AgentThread`

2 **Which component in the Microsoft Agent Framework manages conversation state and stores messages?**

☑ `AgentThread`

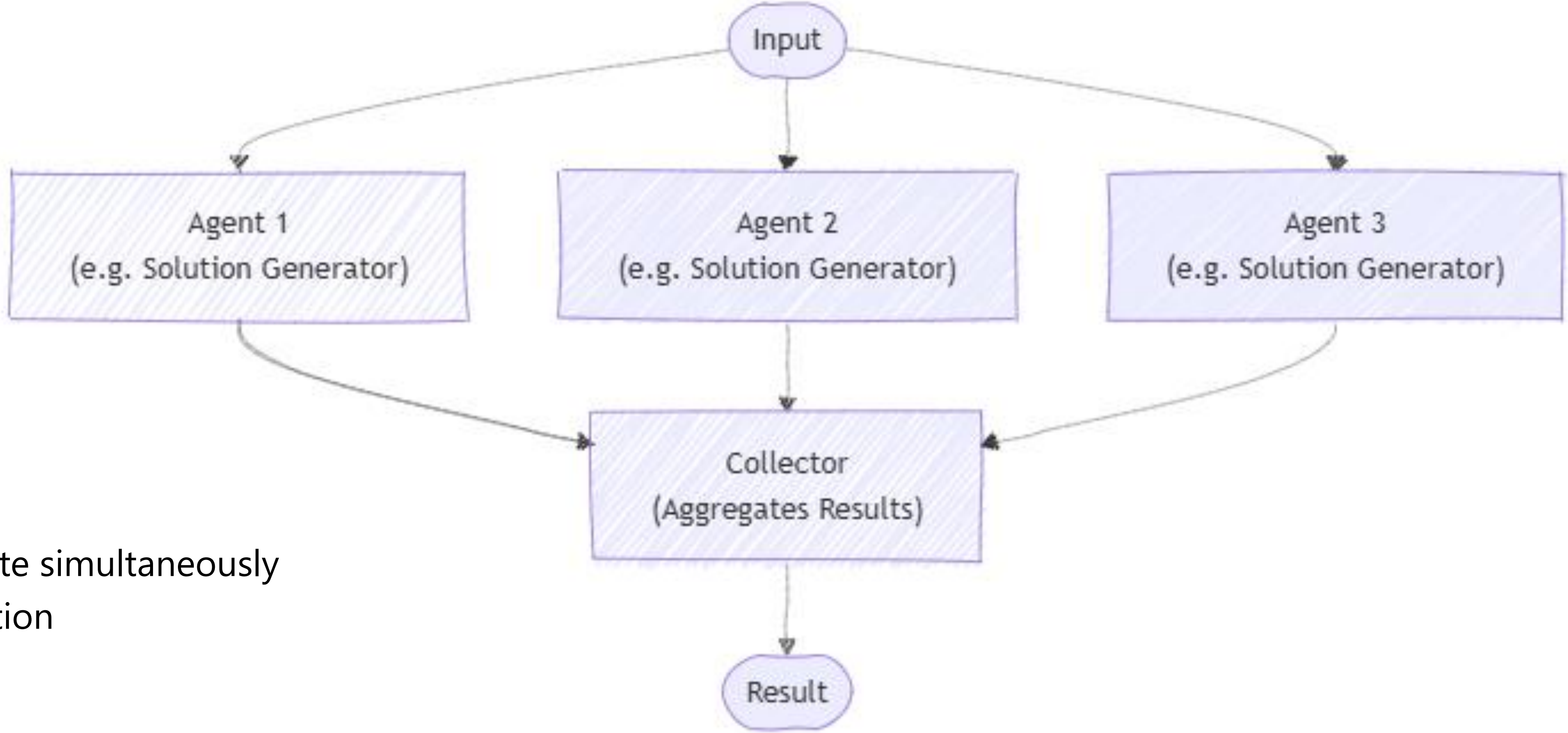
☐ `ChatAgent`

☐ `AzureAIAgentClient`

Orchestrate a multi-agent solution using the Microsoft Agent Framework

https://aka.ms/mslearn-agent-framework-multi-agent

Use concurrent orchestration



Run tasks in parallel

Independent agents execute simultaneously

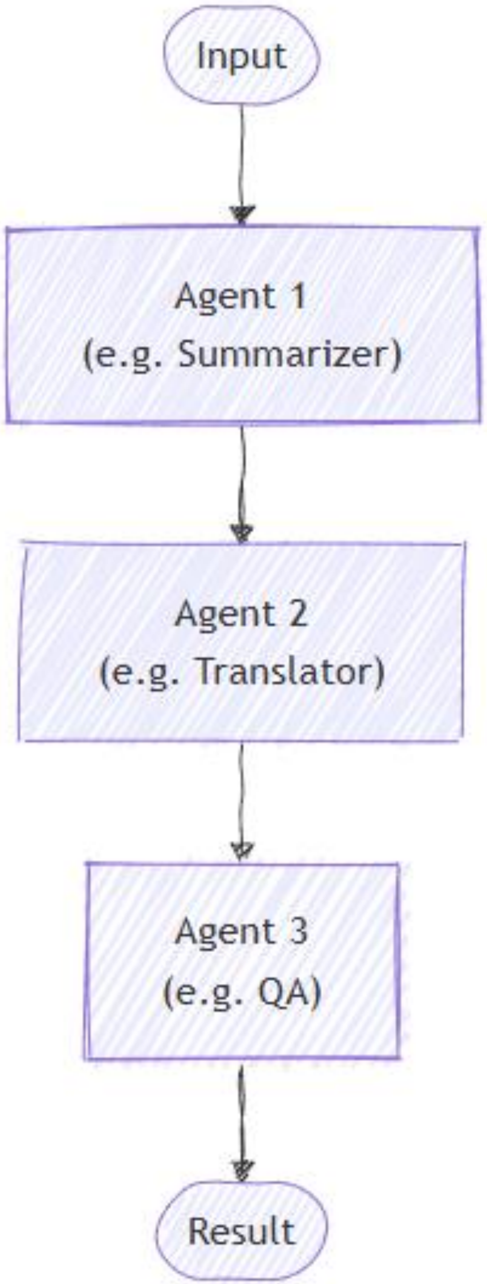
Results merged at completion

Optimized for speed

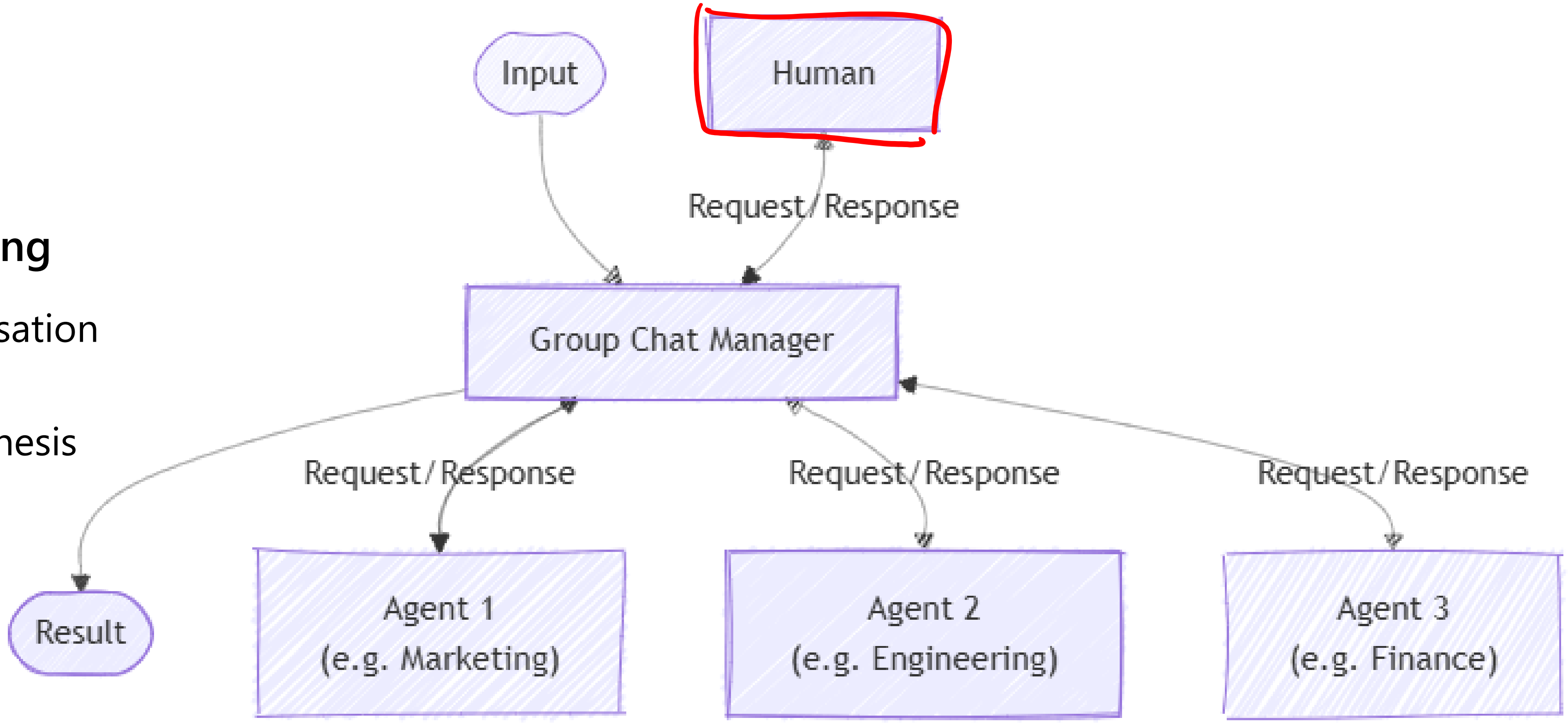
Use sequential orchestration

Step-by-step execution

Each step depends on the previous output
Predictable, ordered flow
Easy to reason about



Use group chat orchestration



Collaborative agent reasoning

Agents share a common conversation
Control shifts dynamically
Best for brainstorming and synthesis

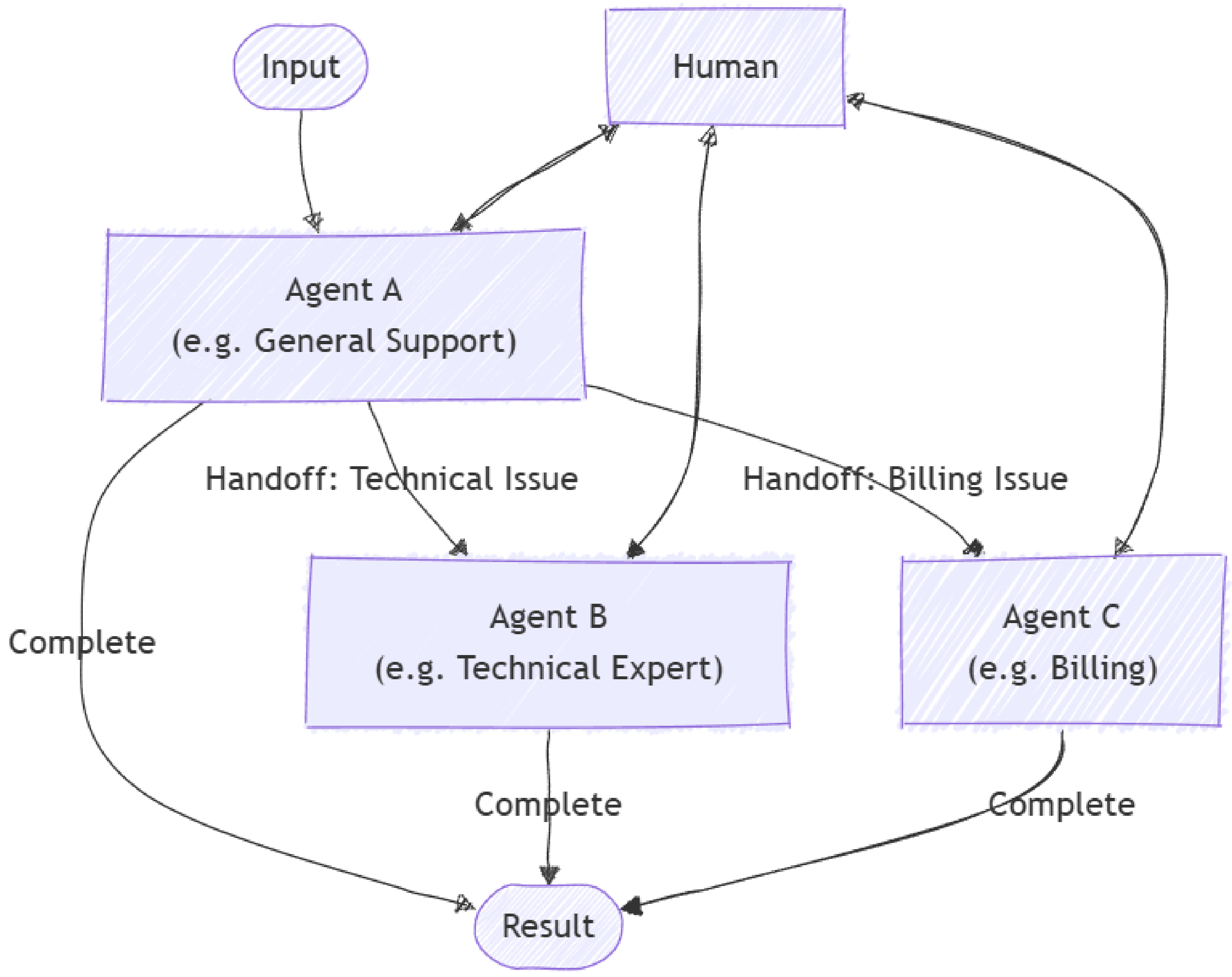
Use handoff orchestration

Delegate by expertise

One agent transfers control to another

Clear ownership per stage

Ideal for role-based workflows



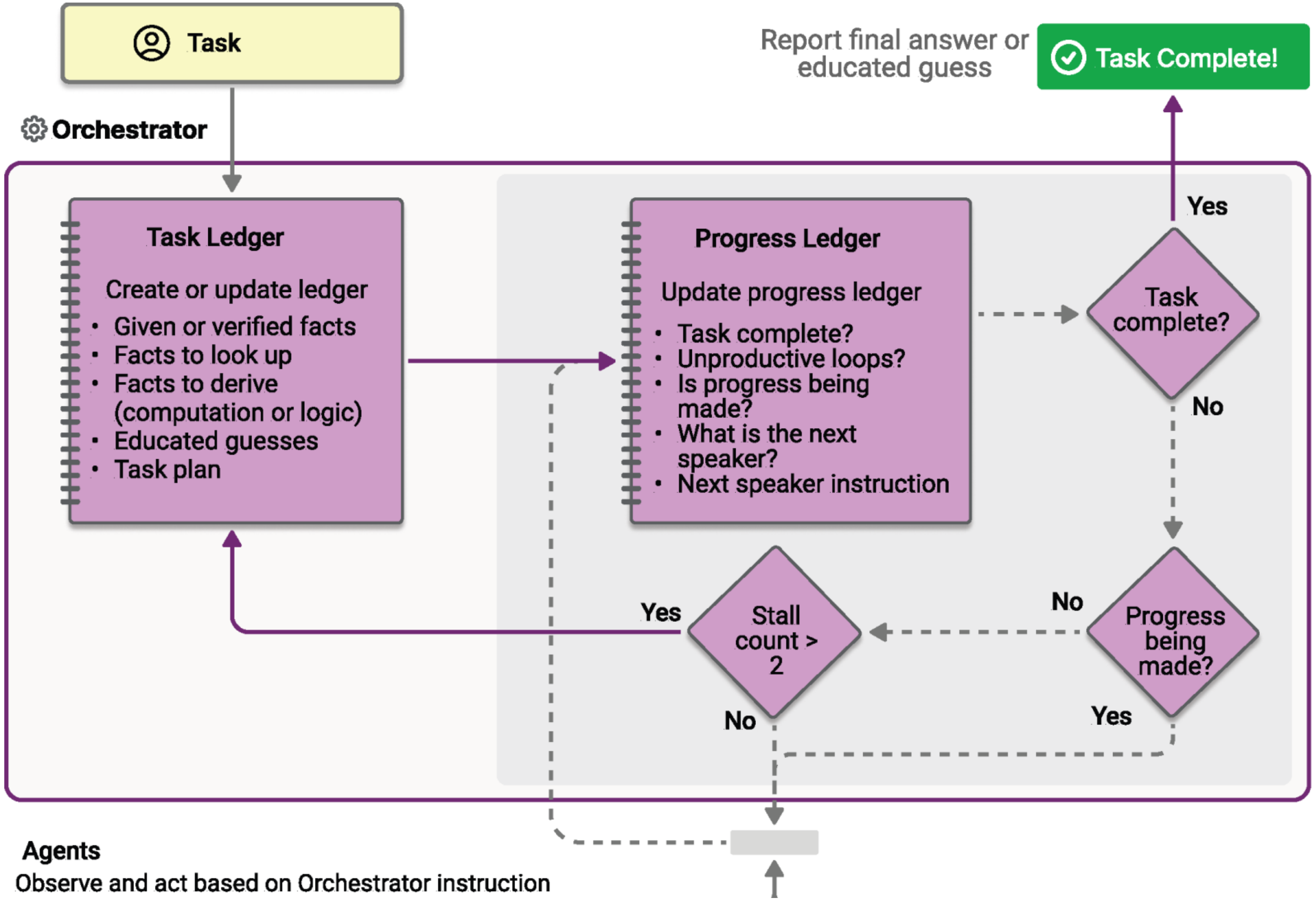
Use Magentic orchestration ?

Adaptive, emergent collaboration

No fixed execution order

Agents self-organize

Suited for complex, open-ended problems



Exercise

Go Deploy

11

13

Workflow Foundry

Multi Agent

Develop a multi-agent solution

In this exercise, you'll:

- Install the Microsoft AI Toolkit VS Code extension
- Deploy a model
- Create AI agents
- Create a sequential orchestration
- Run the app

Start the exercise at:

13

~~12~~. Develop a multi-agent solution with Microsoft Agent Framework

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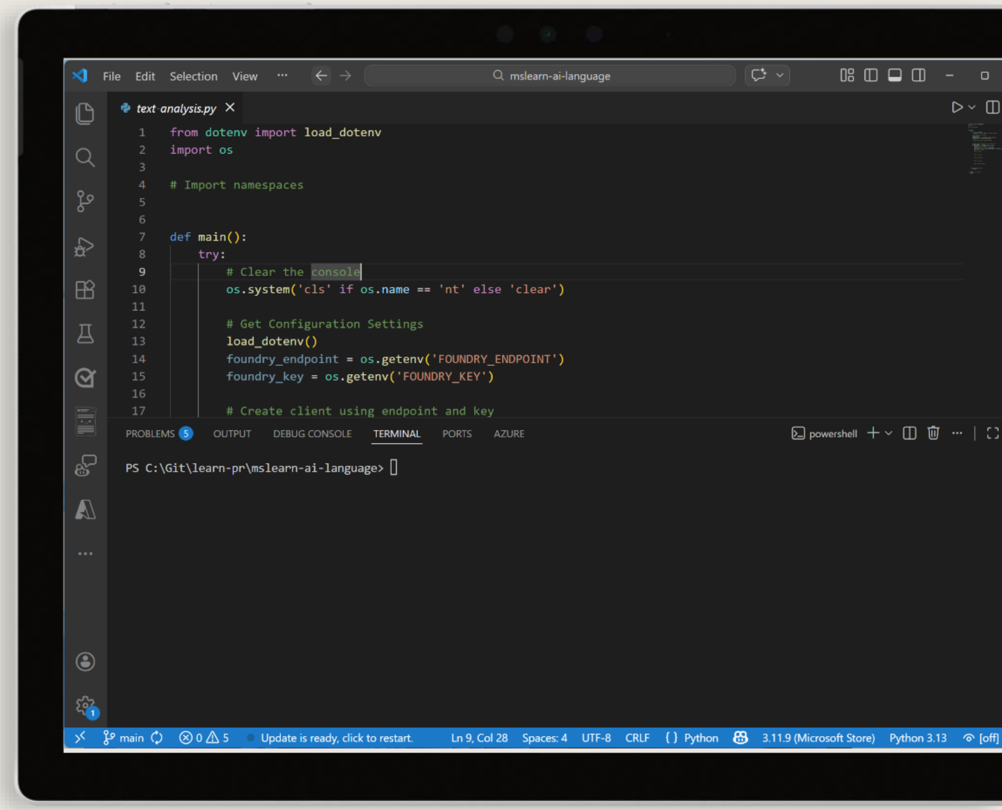

Knowledge check

1 **What's the first step in the Microsoft Agent Framework's unified orchestration workflow?**

☐ Select and create an orchestration pattern
☑ Define your agents and describe their capabilities
☐ Start a runtime to manage execution

2 **For brainstorming and collaborative problem solving among multiple agents, which orchestration pattern is most suitable?**

☑ Group Chat
☐ Magentic
☐ Sequential

Microsoft